



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

January 23, 2026

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Fundamental Definitions

Basic Equations

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Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

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$$\ell \gg \lambda$$

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$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

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- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

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- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla)u)$

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Thank You!

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- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)